

Full Harvest v7.1 - Restoration Guide

This document provides comprehensive instructions for verifying and restoring the Full Harvest v7.1 packages. The harvest includes both source-only and complete repository archives with deterministic builds and comprehensive verification.

Package Overview

Full Harvest v7.1 contains two main archive types:

- `source_only_v7_1.zip` - Complete source code without distribution artifacts (467 files, ~17MB)
- `everything_v7_1.zip` - Complete repository including all distribution files (470 files, ~20MB)

All packages are built deterministically with `SOURCE_DATE_EPOCH=1700000000` for reproducible builds.

Package Contents

Distribution Directory Structure

```
dist/
├─ source_only_v7_1.zip          # Source-only package
├─ everything_v7_1.zip          # Complete repository package
├─ FULL_HARVEST_MANIFEST.json   # Complete file manifest with
hashes
├─ SHA256SUMS                  # SHA256 checksums for
verification
└─ *.parts.json                # Part indexes (if archives
are split)
```

Verification Files

```
DOCS/report/
├─ harvest_filelist.txt        # Complete file tree listing
├─ merkle_root.txt            # Merkle tree root hash
└─ verification_evidence.md    # Build verification logs
```

Quick Start

Option 1: PowerShell (Windows/Cross-platform)

```
# Restore both packages with verification
.\scripts\restore_full_harvest.ps1

# Restore only source package
.\scripts\restore_full_harvest.ps1 -SourceOnly

# Restore to custom directory
.\scripts\restore_full_harvest.ps1 -TargetDir "MyRestore"

# Skip verification (not recommended)
.\scripts\restore_full_harvest.ps1 -Verify:$false
```

Option 2: Bash (Linux/macOS/WSL)

```
# Restore both packages with verification
bash scripts/restore_full_harvest.sh

# Restore only source package
bash scripts/restore_full_harvest.sh --source-only

# Restore to custom directory
bash scripts/restore_full_harvest.sh --target MyRestore

# Skip verification (not recommended)
bash scripts/restore_full_harvest.sh --no-verify
```

Detailed Instructions

Step 1: Pre-restoration Verification

Before extraction, verify package integrity:

```
# Navigate to the distribution directory
cd dist/

# Verify all checksums
sha256sum -c SHA256SUMS
# OR on macOS:
shasum -a 256 -c SHA256SUMS
```

Expected output:

```
source_only_v7_1.zip: OK
everything_v7_1.zip: OK
FULL_HARVEST_MANIFEST.json: OK
```

Step 2: Manual Archive Inspection

Inspect archive contents without extraction:

```
# List source-only archive contents
unzip -l source_only_v7_1.zip | head -20

# List everything archive contents
unzip -l everything_v7_1.zip | head -20

# Check file counts
unzip -l source_only_v7_1.zip | tail -1
unzip -l everything_v7_1.zip | tail -1
```

Step 3: Restoration

Choose your restoration method:

Automated Restoration (Recommended)

PowerShell:

```
# Full restoration with verification
.\scripts\restore_full_harvest.ps1

# Custom options
.\scripts\restore_full_harvest.ps1 -TargetDir "WatchLockAI_v7_1" -
DistDir "./dist"
```

Bash:

```
# Full restoration with verification
bash scripts/restore_full_harvest.sh

# Custom options
bash scripts/restore_full_harvest.sh --target "WatchLockAI_v7_1" --
dist "./dist"
```

Manual Restoration

```
# Create restoration directories
mkdir -p WatchLockAI_Sentinel_Restored_source_only
mkdir -p WatchLockAI_Sentinel_Restored_everything

# Extract source-only package
unzip -q source_only_v7_1.zip -d
WatchLockAI_Sentinel_Restored_source_only/

# Extract everything package
unzip -q everything_v7_1.zip -d
WatchLockAI_Sentinel_Restored_everything/

# Verify extraction
find WatchLockAI_Sentinel_Restored_source_only -type f | wc -l  #
Should be 467
find WatchLockAI_Sentinel_Restored_everything -type f | wc -l  #
Should be 470
```

Advanced Features

Handling Split Archives

For archives exceeding 1GB (automatically split):

1. **Automatic Reassembly:** Both scripts automatically detect and reassemble split archives
2. **Manual Reassembly:**
`` `bash
Check for parts index
ls *.parts.json

```
# Manual reassembly example (if everything_v7_1.zip was split)
cat everything_v7_1.zip.part* > everything_v7_1.zip
` ``
```

Verification Levels

1. Full Verification (default):

- SHA256SUMS verification
- Part integrity checking (if applicable)
- File count validation

2. Quick Verification:

```
bash # Skip SHA256 verification (faster but less secure) bash
scripts/restore_full_harvest.sh --no-verify
```

3. Manual Deep Verification:

```
bash # Compare against manifest python3 -c " import json with
open('FULL_HARVEST_MANIFEST.json') as f: manifest = json.load(f)
print(f'Source          files:          {manifest[\"source_package\"]
[\"files_count\"]}')          print(f'Everything          files:
{manifest[\"everything_package\"][\"files_count\"]}') print(f'Build
timestamp: {manifest[\"build_timestamp\"]}') "
```

Expected Results

Successful Restoration Output

PowerShell:

```
🔧 Full Harvest v7.1 Restoration Script
=====
🔍 Verifying SHA256SUMS...
  ✓ source_only_v7_1.zip
  ✓ everything_v7_1.zip
  ✓ FULL_HARVEST_MANIFEST.json
  ✓ SHA256 verification passed!
📦 Extracting source-only package...
  ✓ Extracted 467 files (XX,XXX,XXX bytes)
📦 Extracting everything package...
  ✓ Extracted 470 files (XX,XXX,XXX bytes)

=====
  ✓ RESTORATION COMPLETE
    Files restored: 937
    Total bytes: XX,XXX,XXX
    Target directories created
      - WatchLockAI_Sentinel_Restored_source_only
      - WatchLockAI_Sentinel_Restored_everything
  ● OK - Full Harvest v7.1 restoration successful!
=====
```

Directory Structure After Restoration

WatchLockAI_Sentinel_Restored_source_only/

- |— DOCS/
- |— tools/
- |— tests/
- |— app_core/
- |— console/
- |— detection/
- |— collectors/
- |— plugins/
- |— service/
- |— ui/
- └─ ... (467 total files)

WatchLockAI_Sentinel_Restored_everything/

- |— (same as above, plus)
- |— dist/
 - |— source_only_v7_1.zip
 - |— everything_v7_1.zip
 - └─ ... (470 total files)

Troubleshooting

Common Issues

1. Permission Denied Errors

```
# Linux/macOS: Add execute permissions
chmod +x scripts/restore_full_harvest.sh

# Alternative: Run with bash directly
bash scripts/restore_full_harvest.sh
```

2. SHA256 Verification Failures

```
# Check file integrity
ls -la dist/
sha256sum dist/source_only_v7_1.zip dist/everything_v7_1.zip

# Compare with expected checksums
cat dist/SHA256SUMS
```

3. Extraction Failures

```
# Check available disk space
df -h .

# Test archive integrity
unzip -t source_only_v7_1.zip
unzip -t everything_v7_1.zip
```

4. Missing Utilities

```
# Install required utilities (Ubuntu/Debian)
sudo apt install unzip sha256sum

# Install required utilities (macOS)
brew install coreutils
```

Recovery Procedures

Complete Re-download

If verification consistently fails:

1. Re-download all harvest files
2. Verify network integrity
3. Check storage device health

Partial Recovery

If only one archive fails:

```
# Restore only the working archive
bash scripts/restore_full_harvest.sh --source-only
# OR
bash scripts/restore_full_harvest.sh --everything
```



Security Considerations

Verification Best Practices

1. **Always verify SHA256SUMS** before extraction
2. **Check file counts** against manifest

3. **Verify restoration environment** is clean
4. **Use official restoration scripts** when possible

Trust Chain

Deterministic Build → SHA256SUMS → Merkle Root → Verification
Evidence
SOURCE_DATE_EPOCH=17000000000 → Reproducible Packages → Verified
Restoration

Integration with VS Code

After successful restoration:

1. **Open in VS Code:**

```
bash code WatchLockAI_Sentinel_Restored_source_only/
```

2. **Verify Python Environment:**

```
bash cd WatchLockAI_Sentinel_Restored_source_only/ python -m  
py_compile $(find . -name "*.py")
```

3. **Check Dependencies:**

```
bash pip install -r requirements.txt pip install -r requirements-  
test.txt
```

Verification Checklist

- ☐ SHA256SUMS verification passed
- ☐ Source package extracted (467 files expected)
- ☐ Everything package extracted (470 files expected)
- ☐ No extraction errors reported
- ☐ Directory structure matches expected layout
- ☐ Python files compile without syntax errors

- [] All restoration scripts executed successfully

Support Information

Restoration Script Issues: Check script help output with `--help` or `-h`

Verification Problems: Examine `DOCS/report/verification_evidence.md`

Build Information: Review `FULL_HARVEST_MANIFEST.json`

Full Harvest v7.1 - Deterministic, Self-Verifying, VS Code Ready

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